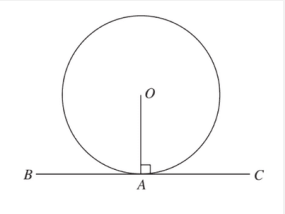
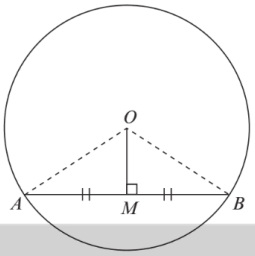
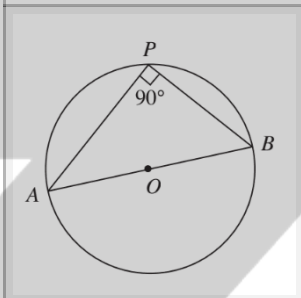
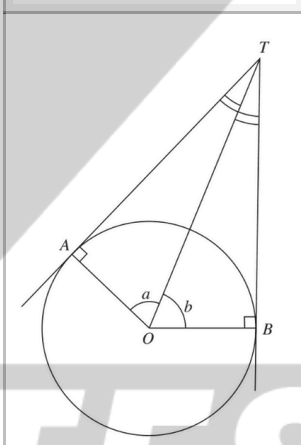
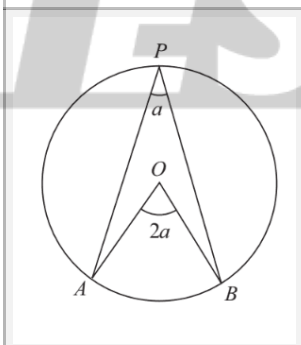
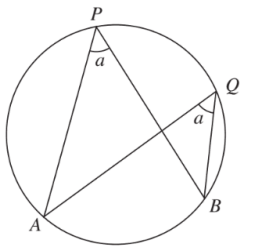
1. Circular measure

1.1. Circle theorems

These theorems **must** be known to solve circular measure questions.

Circle	Theorem
	Radius, OA, forms a 90° degree, or $\frac{\pi}{2}$ radian angle, at the point of contact with a **tangent**.
	The **perpendicular bisector** of a chord, AB, of a circle passes through the centre of the circle, O.
	The **angle in a semi circle** is always equal to 90° degrees, or $\frac{\pi}{2}$ radians.
	Tangents which meet at the **same point** are **equal in length**, $TA = TB$, and angle $a = \text{angle } b$.
	The angle at the centre is **twice** the angle at the circumference.
	Angles in the **same segment** are equal.

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1.2. Introduction to radians

- The radian is the angle formed when the **arc length** is the **same** as the **length** of the **radius**.
  - It is also denoted as a "rad".
- Consider a circle with its radius, r, equal to 1
  - An arc length of 2π would have an angle of 360° degrees, due to the circumference being $2\pi r$.
  - Half of its circumference, or an arc length of π, would have an angle of 180° degrees.

Converting between units

- Using the result, $180^\circ = \pi$, we get these useful ratios
  - 1 radian = $\frac{180^\circ}{\pi}$ degrees
  - 1° degree = $\frac{\pi}{180^\circ}$ radians

These ratios can help as convert from degrees to radians, or vice versa, as shown in the table below.

Converting from	Formula used
Degree to Radian	$\theta \times \frac{\pi}{180^\circ}$
Radian to Degree	$\text{radian} \times \frac{180^\circ}{\pi}$

Example 1

Convert 360° degrees to radians.

Answer

$$360^\circ \times \frac{\pi}{180^\circ} = 2\pi \text{ radians}$$

Example 2

Convert $\frac{\pi}{2}$ radians to degrees.

Answer

$$\frac{\pi}{2} \times \frac{180^\circ}{\pi} = 90^\circ \text{ degrees}$$

1.3. Finding sector area and length

The formulae for finding sector area and sector length can also be used in terms of radians.

Finding sector area

We can find the sector area for a circle using:

Formula in degrees	Formula in radians
Area = $\frac{\theta^\circ}{360^\circ} \times \pi r^2$	Area = $\frac{1}{2}\theta r^2$ [Where θ is in radians]

The formula in terms of radians can be derived easily using the fact $360^\circ = 2\pi$

$$\frac{\theta^\circ}{360^\circ} \times \pi r^2 \implies \frac{\theta^\circ}{2\pi} \times \pi r^2$$

This simplifies to the formula in radians

$$\text{Area} = \frac{1}{2}\theta r^2 \text{ [Where } \theta \text{ is in radians]}$$

Finding sector length

We can find the sector length for a circle using:

Formula in degrees	Formula in radians
Length = $\frac{\theta^\circ}{360^\circ} \times 2\pi r$	Length = θr [Where θ is in radians]

The formula in terms of radians can be derived easily using the fact $360^\circ = 2\pi$

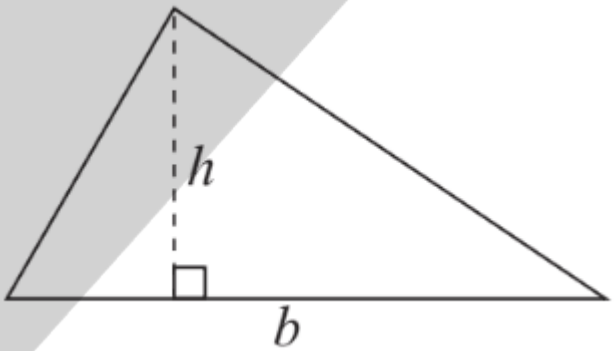
$$\frac{\theta^\circ}{360^\circ} \times 2\pi r \implies \frac{\theta^\circ}{2\pi} \times 2\pi r$$

This simplifies to the formula in radians

$$\text{Length} = \theta r \text{ [Where } \theta \text{ is in radians]}$$

1.4. Area of a triangle

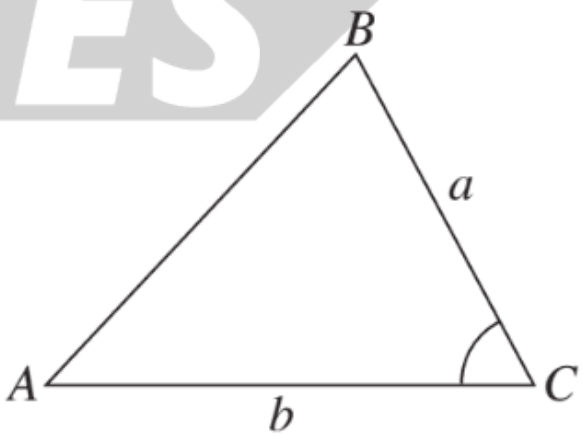
Let us take a triangle, where h represents the height and b represents the base.



- The area of this triangle can be found using the formula $\frac{1}{2} \times base \times height$ or $\frac{1}{2} \times b \times h$.

Alternative formula for the area of a triangle

Consider another triangle, where you aren't given the base but have the values of 2 sides and 1 angle.



- To find the area of this triangle, we must first draw a perpendicular from B to the line AC

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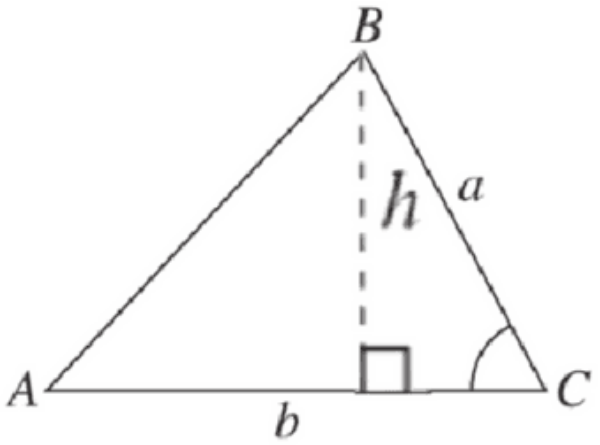

Utilising trigonometry, we observe that

$$sin(C) = \frac{h}{a} \implies h = sin(C) \times a$$

Taking the base as b and height as $sin(C) \times a$, we can substitute it into the formula to find the area of this triangle as

$$\frac{1}{2} \times a \times b \times sin(C)$$

- Note that this formula requires you to use an angle C, such that it is not facing the side a or side b as shown in the diagram above.

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